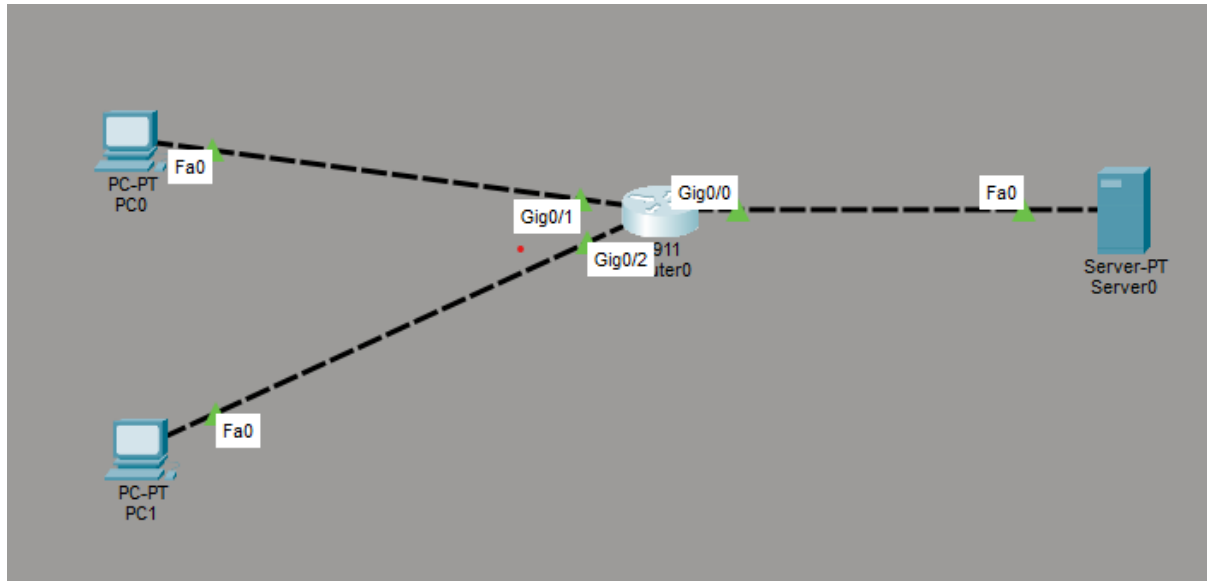


Nama: Firdho Aulia

NIM : 2410130007

1. Topologi



Daftar Ip :

PC 0 : 192.168.2.2, Gateway : 192.168.2.1

PC 1 : 192.168.3.2, Gateway : 192.168.3.1

Server : 192.168.1.2, Gateway : 192.168.1.1

2. Router IPS

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip ips config location flash:
Router(config)#ip ips name SECURE
Router(config)#int g0/0
Router(config-if)#ip ips SECURE in
Router(config-if)#
%IPS-6-ENGINE_BUILDS_STARTED: 00:09:05 UTC Mar 01 1993

%IPS-6-ENGINE_BUILDING: atomic-ip - 3 signatures - 1 of 13 engines

%IPS-6-ENGINE_READY: atomic-ip - build time 8 ms - packets for this engine will be scanned

%IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed time 8 ms

Router(config-if)#exit
Router(config)#int g0/1
Router(config-if)#ip ips SECURE in
Router(config-if)#exit
Router(config)#int g0/2
Router(config-if)#ip ips SECURE in
Router(config-if)#exit
```

IPS tidak hanya di pasang di jalur menuju server tapi juga ke jalur PC client karena saya mengadopsi strategi “Zero Trust”

3. Konfigurasi ip IPS

```
ip ips config location flash: retries 1
ip ips name SECURE
!
!
!
!
!
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 ip ips SECURE in
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 192.168.2.1 255.255.255.0
 ip ips SECURE in
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 ip address 192.168.3.1 255.255.255.0
 ip ips SECURE in
 duplex auto
 speed auto
```

Ip sudah di set dengan nama ips SECURE

4. Pembuatan Rule Block / Scanning dengan signature 2004

```
Router(config)#ip ips signature-definition
Router(config-sigdef)#signature 2004 0
Router(config-sigdef-sig)#status
Router(config-sigdef-sig-status)#enabled true
Router(config-sigdef-sig-status)#retired false
Router(config-sigdef-sig-status)#exit
Router(config-sigdef-sig)#engine
Router(config-sigdef-sig-engine)#event-action deny-packet-inline
Router(config-sigdef-sig-engine)#exit
Router(config-sigdef-sig)#exit
Router(config-sigdef)#exit
Do you want to accept these changes? [confirm]
*IPS-6-ENGINE_BUILDS_STARTED:
*IPS-6-ENGINE_BUILDING: atomic-ip - 303 signatures - 3 of 13 engines
*IPS-6-ENGINE_READY: atomic-ip - build time 480 ms - packets for this engine will be scanned
*IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed time 648 ms
```

5. Pengujian dari PC ke server :

Sebelum set IPS

```
Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time=2ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

Sesudah set IPS

```
Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127
Reply from 192.168.1.2: bytes=32 time=2ms TTL=127
Reply from 192.168.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

Masih boleh karena IPS hanya mengecek trafik normal atau tidak (jika normal = aman)

Setelah set IPS + Rule Blocking / Scanning

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Catatan

Sebenarnya IPS memiliki sistem untuk memblokir jika ping terdeteksi sebagai gangguan seperti melakukan Flooding atau Scanning, namun karena keterbatasan aplikasi di Cisco saya pembedaan antara trafik normal dan gangguan tidak bisa dilakukan karena Threshold dan interval tidak bisa di set menyebabkan setelah di lakukannya blocking rule maka ping ke server akan di block walaupun ping normal.

Bukti blockir dari sistem

```
Router(config)#show logging
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 32 messages logged, xml disabled,
filtering disabled
Monitor logging: level debugging, 32 messages logged, xml disabled,
filtering disabled
Buffer logging: level debugging, 0 messages logged, xml disabled,
filtering disabled

Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

ESM: 0 messages dropped
Trap logging: level informational, 32 message lines logged

Log Buffer (4096 bytes):

%SYS-5-LOG_CONFIG_CHANGE: Buffer logging: level debugging, xml disabled, filtering disabled, size (4096)
%SYS-5-LOG_CONFIG_CHANGE: Buffer logging: level debugging, xml disabled, filtering disabled, size (4096)

%IPS-4-SIGNATURE: Sig:2004 Subsig:0 Sev:25 [192.168.2.2 -> 192.168.1.2:0] RiskRating:25
%IPS-4-SIGNATURE: Sig:2004 Subsig:0 Sev:25 [192.168.2.2 -> 192.168.1.2:0] RiskRating:25
%IPS-4-SIGNATURE: Sig:2004 Subsig:0 Sev:25 [192.168.2.2 -> 192.168.1.2:0] RiskRating:25
%IPS-4-SIGNATURE: Sig:2004 Subsig:0 Sev:25 [192.168.2.2 -> 192.168.1.2:0] RiskRating:25
%IPS-4-SIGNATURE: Sig:2004 Subsig:0 Sev:25 [192.168.2.2 -> 192.168.1.2:0] RiskRating:25
%IPS-4-SIGNATURE: Sig:2004 Subsig:0 Sev:25 [192.168.2.2 -> 192.168.1.2:0] RiskRating:25
%IPS-4-SIGNATURE: Sig:2004 Subsig:0 Sev:25 [192.168.2.2 -> 192.168.1.2:0] RiskRating:25
%IPS-4-SIGNATURE: Sig:2004 Subsig:0 Sev:25 [192.168.2.2 -> 192.168.1.2:0] RiskRating:25
%SYS-5-CONFIG_I: Configured from console by console
```

Kesimpulan

IPS bekerja dengan konsep "apa yang anda lakukan" (berbasis perilaku dan *signature*), sedangkan **Firewall/ACL bekerja dengan konsep "siapa anda"** (berbasis identitas IP). Melalui aktivasi Signature 2004, sistem tidak lagi sekadar memfilter alamat pengirim, melainkan melakukan inspeksi mendalam untuk mendeteksi pola gangguan seperti *scanning* atau *flooding*; meskipun simulator memiliki keterbatasan dalam mengatur ambang batas (*threshold*), bukti log %IPS-4-SIGNATURE telah memverifikasi bahwa router mampu secara proaktif menghentikan aktivitas berbahaya berdasarkan analisis tindakan trafik secara *real-time*.